
When Do Language Models Beat Crowd Forecasts? A Matched Comparison Across Long-Horizon Sparse and Short-Horizon Dense Regimes

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Abstract

Forecasting systems are only useful in practice if we know when they can beat existing human judgment. We study that question for binary event forecasting by comparing GPT-4.1 against contemporaneous human crowd forecasts under two information regimes derived from the same local dataset: a LONG-TERM LOW-DATA regime with sparse historical signal and a SHORT-TERM HIGH-DATA regime with dense historical signal close to resolution. We evaluate 60 resolved questions, sample 30 cutoff timepoints per regime, and elicit model forecasts under two prompt conditions: QUESTION-ONLY and QUESTION+HISTORY. This design isolates both information regime and access to structured crowd trajectories while keeping the task, dataset, and outcome space fixed.

The results are mixed. In the LONG-TERM LOW-DATA regime, GPT-4.1 outperforms the crowd in the QUESTION-ONLY condition, improving mean Brier from 0.200 to 0.151 and accuracy from 0.600 to 0.767. In the SHORT-TERM HIGH-DATA regime, the same model underperforms the crowd without trajectory information, with Brier 0.131 versus 0.065. Adding structured historical crowd data closes that gap and slightly reverses it, yielding Brier 0.056 for the model versus 0.065 for the crowd. None of the paired Brier differences reach $p < 0.05$ at $n = 30$ per condition, so we treat the effect directions as suggestive rather than definitive.

Our main conclusion is narrower than a blanket claim of LLM superiority. Text-only LLM forecasts appear competitive in long-horizon sparse settings, while short-horizon performance depends strongly on access to structured historical forecasting data.

1 Introduction

Forecasts guide decisions in policy, business, and operations, so the practical question is not whether language models can generate plausible probabilities, but when they can outperform existing human judgment. For deployment, that distinction matters: a system that helps on sparse early-stage questions may be useful even if it fails near resolution, while a system that only succeeds after seeing crowd trajectories may be better viewed as a synthesizer than an independent forecaster.

Recent work has made this picture harder to read. On one hand, retrieval-augmented forecasting systems built on language models have approached human crowd performance on real forecasting-platform questions Halawi et al. [2024]. On the other hand, direct comparisons between general-purpose LLMs and human crowds have reported large gaps in favor of humans Schoenegger and Park [2023]. A separate line of work studies numeric time-series forecasting with LLMs and often reports strong benchmark results, but these studies usually do not include human baselines and often evaluate a different problem altogether Gruver et al. [2023], Jin et al. [2024], Tan et al. [2024]. As

a result, the simple hypothesis that “LLMs outperform humans in both short-term high-data and long-term low-data forecasting” remains under-tested.

What is missing? Existing evidence often mixes datasets, tasks, or information conditions. Event forecasting papers compare to humans but focus on open-world binary questions; time-series papers study dense numeric sequences without human crowd baselines; and positive and negative findings are rarely tested within one controlled dataset. That makes it difficult to tell whether performance differences come from horizon, information density, prompt design, or task mismatch.

Our approach. We address this gap with a matched within-dataset comparison. Using one local binary-question forecasting dataset, we construct two regimes from the same pool of resolved questions: LONG-TERM LOW-DATA, defined by at least 60 days to resolution and at most five crowd predictions observed, and SHORT-TERM HIGH-DATA, defined by at most 14 days to resolution and at least 30 crowd predictions observed. We then compare GPT-4.1, the contemporaneous human crowd forecast, and a 0.5 baseline at the same historical cutoff. We further run two prompt conditions: QUESTION-ONLY, which uses only question context, and QUESTION+HISTORY, which adds a structured summary of the crowd trajectory available before the cutoff.

What do we find? The answer is mixed and regime-dependent. In LONG-TERM LOW-DATA, the QUESTION-ONLY model forecast improves mean BRIER from 0.200 for the crowd to 0.151 for the LLM, a paired mean difference of -0.050 . In SHORT-TERM HIGH-DATA, the same prompt underperforms badly, with BRIER 0.131 versus 0.065 for the crowd. Adding historical crowd structure nearly closes that gap and slightly reverses it, bringing model BRIER to 0.056. None of these paired differences are statistically significant at $\alpha = 0.05$, but the directional pattern is consistent with the view that LLMs help more with sparse long-horizon reasoning than with late-stage consensus tracking.

- We construct a controlled within-dataset evaluation that compares LLM and crowd forecasts across matched sparse and dense information regimes.
- We conduct a prompt ablation that separates text-only forecasting from history-conditioned synthesis of human trajectory information.
- We report paired BRIER comparisons, bootstrap confidence intervals, and calibration plots that clarify where the model gains and where it fails.
- We provide a narrower empirical conclusion: LLM competitiveness depends strongly on information regime and access to historical forecast structure.

The rest of the paper is organized as follows. Section 2 situates the study relative to event-forecasting and time-series work. Section 3 describes the dataset construction, prompting conditions, and statistical analysis. Section 4 reports the main results and ablations, and Section 5 discusses mechanisms, limitations, and implications.

2 Related Work

LLMs for human-comparable event forecasting. The most relevant line of work studies binary event forecasting on real questions with human crowd baselines. Halawi et al. develop a retrieval-augmented language-model forecasting pipeline and show that it approaches, but does not generally exceed, human crowd performance on a large curated benchmark from multiple forecasting platforms Halawi et al. [2024]. Their strongest gains appear when the crowd is uncertain or when forecasts are made earlier in a question’s lifecycle. Schoenegger and Park report a more negative result: GPT-4 underperforms a live Metaculus crowd and does not significantly beat a 0.5 baseline in a tournament-style evaluation Schoenegger and Park [2023]. Paleka et al. focus on logical consistency checks for language-model forecasters rather than direct human comparison, but their framework is relevant because it offers a way to diagnose forecast quality before outcomes resolve Paleka et al. [2024]. Our work sits closest to this literature. Unlike large retrieval systems, we use a controlled local-context setup, and unlike live-tournament studies, we vary information regime within one dataset.

LLMs for numeric time-series forecasting. A separate literature treats forecasting as sequence extrapolation over dense numeric observations. Gruver et al. show that general language models can perform zero-shot time-series forecasting by serializing numeric sequences into text Gruver et al.

[2023]. Jin et al. propose Time-LLM, which reprograms frozen LLMs for time-series forecasting through patch embeddings and prompt-style conditioning Jin et al. [2024]. Liu et al. extend this direction with large time-series foundation models trained directly for temporal tasks Liu et al. [2024], and Guo et al. argue that temporal distillation can further improve LLM-based forecasting Guo et al. [2026]. These papers are important evidence that LLM-style architectures can work well on high-data forecasting problems, but they do not answer whether such systems outperform human crowd judgments on event questions.

Skeptical ablations and task-definition concerns. Later analyses caution against attributing time-series gains too broadly to language modeling itself. Tan et al. reproduce several influential LLM time-series systems and find that simpler ablations often match or exceed them, suggesting that gains may come from patching or attention structure rather than language transfer Tan et al. [2024]. This critique matters for our question because the user hypothesis mixes two forecasting settings that behave differently in the literature. We therefore stay within binary question forecasting and vary information regime directly instead of combining event forecasting with dense numeric benchmarks. That choice makes our comparison narrower, but it gives a cleaner answer to when an LLM can match or beat a human crowd on the same underlying task.

3 Methodology

Task and dataset. We study resolved binary forecasting questions from the local ‘forecast_questions_platforms’ test split. Each item includes question text, resolution criteria, background context, resolution date, and a trajectory of crowd forecasts over time. Rather than compare across datasets, we convert this single source into matched question-timepoints that differ in horizon and available historical signal.

Regime construction. We define two operational regimes. The LONG-TERM LOW-DATA regime requires a forecast cutoff at least 60 days before resolution with at most five crowd predictions observed before the cutoff. The SHORT-TERM HIGH-DATA regime requires a cutoff at most 14 days before resolution with at least 30 crowd predictions observed. We sample 30 items per regime, enforce no question overlap between the short-horizon and long-horizon samples, and evaluate two prompt conditions per item. The final sample contains 60 unique question-timepoints and 120 LLM forecasts. The sampled LONG-TERM LOW-DATA items are genuinely sparse, with mean 119.2 days to resolution and mean 1.0 crowd predictions observed. The sampled SHORT-TERM HIGH-DATA items are dense and late-stage, with mean 1.5 days to resolution and mean 100.0 crowd predictions observed.

Prompt conditions. We use GPT-4.1 with temperature 0.0 and JSON-only output parsing. In the QUESTION-ONLY condition, the model sees the question, dates, background, and resolution criteria. In the QUESTION+HISTORY condition, it also sees a structured summary of the crowd trajectory available before the cutoff. This comparison is central to the paper: QUESTION-ONLY measures how well the model forecasts from question context alone, while QUESTION+HISTORY measures how well it can synthesize structured historical human signals. We use the same local prompt template for both conditions, with the history block added only in the second condition.

Baselines and metrics. We compare the LLM against two baselines at the same cutoff: the contemporaneous human crowd probability and an uninformed 0.5 forecast. Our primary metric is BRIER score, the standard squared-error metric for binary forecasting. We also report thresholded accuracy and qualitative calibration plots. Lower BRIER is better. Because the goal is to compare forecasters item by item, we compute paired differences in BRIER for each question-timepoint.

Statistical analysis. For each regime and prompt condition, we run a Wilcoxon signed-rank test on paired LLM-versus-crowd BRIER scores and compute a bootstrap 95% confidence interval for the mean paired difference (LLM – Crowd). This design tests whether the LLM systematically improves on the crowd rather than only shifting aggregate averages. We treat each sampled question-timepoint as the unit of analysis and use $\alpha = 0.05$.

Implementation details. The experiment runs under Python 3.12.8 with ‘openai’ 2.38.0, ‘numpy’ 2.4.6, and ‘pandas’ 3.0.3. The machine exposes four NVIDIA RTX A6000 GPUs, but they are unused because the evaluation is API-based. Total token usage is 95,148 across both conditions, and the approximate API cost under the stated pricing assumption is \$0.22.

Table 1: Observed characteristics of the sampled evaluation items. The two regimes differ sharply in both horizon and available historical crowd signal, which is the intended contrast in the experiment.

Regime	Avg. Days	Median Days	Avg. Crowd Preds.	Median Preds.
LONG-TERM LOW-DATA	119.2	125.0	1.0	1.0
SHORT-TERM HIGH-DATA	1.5	1.0	100.0	101.0

Table 2: Paired LLM-versus-crowd BRIER analysis. Negative mean differences favor the LLM. None of the comparisons are statistically significant at $\alpha = 0.05$.

Regime	Prompt	Wilcoxon p -value	Mean Diff. (LLM – Crowd)	95% CI
LONG-TERM LOW-DATA	QUESTION-ONLY	0.309	-0.050	[-0.143, 0.037]
LONG-TERM LOW-DATA	QUESTION+HISTORY	0.889	-0.026	[-0.078, 0.018]
SHORT-TERM HIGH-DATA	QUESTION-ONLY	0.943	+0.066	[-0.012, 0.154]
SHORT-TERM HIGH-DATA	QUESTION+HISTORY	0.239	-0.010	[-0.031, 0.001]

4 Results

Main comparison. Table 3 reports the core results. The strongest text-only result appears in LONG-TERM LOW-DATA: GPT-4.1 improves mean BRIER from 0.200 for the crowd to 0.151 and raises accuracy from 0.600 to 0.767. The same prompt fails in SHORT-TERM HIGH-DATA, where the crowd remains much better with BRIER 0.065 versus 0.131 for the model. This reversal is important because it rejects the broad claim that a single text-only LLM forecast dominates the crowd in both regimes.

Effect of historical crowd trajectories. Adding structured crowd history has opposite effects across regimes. In SHORT-TERM HIGH-DATA, it sharply improves the model from 0.131 to 0.056 BRIER, narrowly beating the crowd’s 0.065 and raising accuracy from 0.833 to 0.933. In LONG-TERM LOW-DATA, the same addition worsens the model from 0.151 to 0.174 BRIER. This pattern suggests that late-stage dense questions benefit from explicit temporal human signal, while early sparse questions may be harmed by anchoring to a thin or noisy trajectory.

Statistical tests. Table 2 shows the paired analysis. None of the four LLM-versus-crowd comparisons reach $p < 0.05$. The most favorable mean paired difference is -0.050 in LONG-TERM LOW-DATA under QUESTION-ONLY, but its 95% confidence interval $[-0.143, 0.037]$ crosses zero. The least favorable is $+0.066$ in SHORT-TERM HIGH-DATA under QUESTION-ONLY, with confidence interval $[-0.012, 0.154]$. We therefore interpret direction, magnitude, and consistency across conditions as suggestive evidence rather than proof of superiority.

Error analysis. The qualitative failures match the aggregate pattern. In LONG-TERM LOW-DATA, several extreme crowd probabilities on eventual NO outcomes create room for the LLM to win with more moderate forecasts. The report highlights a labor-force-participation question where the crowd assigned 0.98 to an outcome that resolved NO, while the LLM assigned 0.45 and incurred far less loss. A similar pattern appears on a Serbia/Kosovo conflict question, where the crowd was at 0.70 and the LLM at 0.18 before a NO resolution. In SHORT-TERM HIGH-DATA without history, the opposite happens: the model stays near 0.05–0.15 on several eventual YES outcomes that the crowd had already updated toward, including a Biden impeachment inquiry and Spain winning the 2023 Women’s World Cup. Those misses mostly disappear once the model sees the historical trajectory.

5 Discussion

What do the regime differences mean? The simplest interpretation is that the model and the crowd rely on different sources of signal. In LONG-TERM LOW-DATA, where only about one crowd prediction is visible on average and more than 100 days remain before resolution, broad world knowledge and textual reasoning may be more useful than crowd momentum. In SHORT-TERM HIGH-DATA, the key signal is often not in the question text itself but in the sequence of recent updates that encode late-breaking evidence and collective reaction. That is exactly the signal the QUESTION-ONLY model lacks and the QUESTION+HISTORY model recovers.

Table 3: Main forecasting results by regime and prompt condition. Lower BRIER is better. Best LLM-or-crowd result within each row is shown in **bold**.

Regime	Prompt	n	LLM BRIER	Crowd BRIER	0.5 Baseline	LLM Acc.	Crowd Acc.	Mean Diff.
LONG-TERM LOW-DATA	QUESTION-ONLY	30	0.151	0.200	0.250	0.767	0.600	-0.050
LONG-TERM LOW-DATA	QUESTION+HISTORY	30	0.174	0.200	0.250	0.633	0.600	-0.026
SHORT-TERM HIGH-DATA	QUESTION-ONLY	30	0.131	0.065	0.250	0.833	0.900	+0.066
SHORT-TERM HIGH-DATA	QUESTION+HISTORY	30	0.056	0.065	0.250	0.933	0.900	-0.010

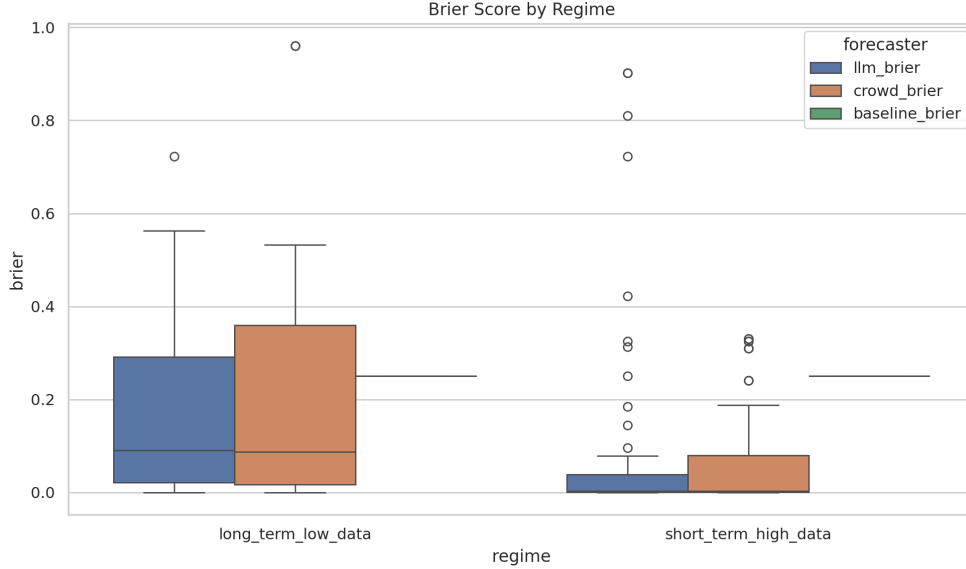


Figure 1: Mean BRIER by regime and prompt condition. The QUESTION-ONLY model performs best in LONG-TERM LOW-DATA, while the QUESTION+HISTORY condition is necessary to match or slightly beat the crowd in SHORT-TERM HIGH-DATA.

What does this imply for LLM deployment? Our results argue against treating LLMs as uniform replacements for human forecasting crowds. A text-only LLM may be useful as an early-stage independent forecast or as a complement when crowd information is sparse. By contrast, late-stage deployment appears to require structured external signal. In this setup, crowd trajectory summaries provide that signal, but retrieval at the historical cutoff could play a similar role in future work. The practical lesson is that system design should depend on information regime, not on a single aggregate claim about model strength.

How does this relate to prior work? The results are consistent with the literature review. They align with Halawi et al.’s finding that language models can be competitive when the crowd is uncertain or when forecasts are made earlier Halawi et al. [2024], and they remain compatible with Schoenegger and Park’s negative result for direct human comparison in realistic forecasting settings Schoenegger and Park [2023]. Our study adds a controlled within-dataset lens: both patterns can be true at once, depending on horizon and information density.

Limitations. The study is small, with only 30 items per regime and prompt condition, and none of the paired differences are statistically significant at $\alpha = 0.05$. We do not use live retrieval, so the LLM only sees local dataset context rather than the open-web evidence that a real forecaster might consult. The QUESTION+HISTORY condition is not a pure independent forecast because it exposes human trajectory information directly to the model. The LONG-TERM LOW-DATA regime is also an especially strict operationalization of low-data, with a mean of only one observed crowd prediction. Finally, the SHORT-TERM HIGH-DATA sample is dominated by Metaculus questions, which may limit cross-platform generalization.

Broader implications. Forecasting papers often compress results into a single headline about whether LLMs beat humans. Our results suggest that this framing is too coarse. What matters is which information is available at forecast time and whether the model can access the signals that

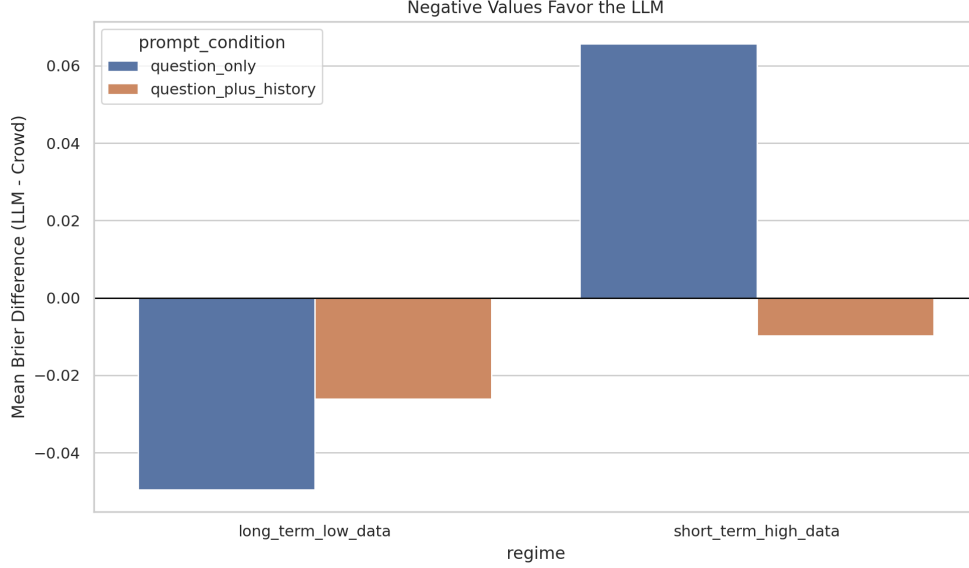


Figure 2: Paired mean BRIER difference (LLM – Crowd) by regime and prompt. Negative values favor the LLM. The sign flips across regimes for QUESTION-ONLY, and history conditioning moves the short-horizon result close to parity.

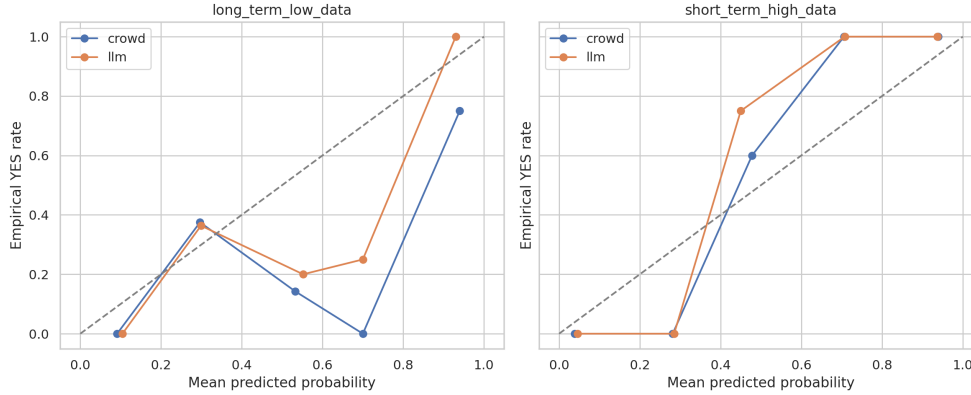


Figure 3: Calibration under the QUESTION+HISTORY condition. The history-conditioned model is reasonably aligned overall, but the main improvement comes from correcting large short-horizon misses rather than from a uniform calibration gain across all bins.

humans already aggregate. A more useful research direction is to characterize the division of labor between LLMs, retrieval, and human crowds rather than search for a single overall winner.

6 Conclusion

We examined whether GPT-4.1 outperforms contemporaneous human crowd forecasts in two matched regimes derived from the same binary-question dataset: LONG-TERM LOW-DATA and SHORT-TERM HIGH-DATA. The answer is mixed. In the sparse long-horizon regime, the text-only model is directionally better than the crowd, improving mean BRIER from 0.200 to 0.151. In the dense short-horizon regime, the text-only model is much worse than the crowd, but adding structured historical trajectory information improves it to near parity and a slight mean advantage. None of the paired differences are statistically significant with $n = 30$ per condition, so the evidence is suggestive rather than definitive.

Our main contribution is a controlled within-dataset comparison that separates independent text-based forecasting from history-conditioned synthesis. The key takeaway is that LLM competitiveness depends strongly on information regime: question context alone can be enough for some long-horizon sparse forecasts, while short-horizon performance appears to require structured historical signal.

The next step is to test whether live retrieval at the historical cutoff can close the short-horizon gap without directly exposing crowd probabilities. Larger samples, more platforms, and comparisons to newer OpenAI and non-OpenAI models would also help determine how robust these regime-specific patterns are.

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